

(e.g., “quant-timental” managers) information to construct portfolios based on what stocks are likely to outperform their peer group over time. These managers often have a long term view on the stocks.

The arrival price benchmark is also an appropriate benchmark price for situations where a market event triggers the portfolio manager or trader to release an order to the market.

The arrival price benchmark is:

$$E_0[\text{Arrival Cost}] = (\bar{P} - P_0) \cdot \text{Side}$$

The E_0 notation here is used to denote that this is the expected cost at the beginning of trading. The expected cost for a buy order with strategy expressed in terms of trading rate is:

Let,

$$P_0 = \text{Arrival Price}$$

$$\bar{P} = P_0 + (b_1 \cdot I' \cdot \alpha) + (1 - b_1) \cdot I'$$

Then the expected cost is:

$$E_0[\bar{P} - P_0] = (b_1 \cdot I' \cdot \alpha) + (1 - b_1) \cdot I' = MI'$$

And we have,

$$E_0[\text{Arrival Cost}] = (b_1 \cdot I' \cdot \alpha) + (1 - b_1) \cdot I' \quad (8.9)$$

For the arrival price benchmark the expected cost is equal to the market impact of the trade.

Historical Price Benchmark

Quantitative managers who run optimization models may select a historical price as their benchmark if this represents the price used in the optimization process. Until recently, many quant managers would run optimizers after the close incorporating the closing price on the day. Optimizers determine the mix of stocks and shares to hold in the portfolio and the corresponding trade list for the next morning. These orders are then submitted to the market at the open the next day. The overnight price movement represents a price jump or discontinuity in the market that the trader is not able to participate with and represents either a sunk cost or a saving to the fund at the time trading begins. If the manager is looking to buy shares in a stock that closed at \$30.00 but opened at \$30.05, the \$0.05/share move represents a sunk cost to the manager. But